Monthly Collection Cycle — Workflow & Algorithm

Overview

A five-phase end-to-end process from cycle open to settlement, covering contribution collection, late fee tiers, bank reconciliation, and reporting.

Account Structure

Account	Purpose
Master Bank Account	Cash from bank statement imports
Master Cash Account	Cash currently held by the system
Master Fund Account	Total value of the fund
Member Cash Account	Cash belonging to the member
Member Fund Account	Member's contributions to the fund

Master Account Invariant (assert nightly): `- Master Fund Account = Σ(all Member Fund Accounts)` - `Master Cash Account = Σ(all Member Cash Accounts)`

Phase 1 — Cycle Initialisation (Day 1)

1. **Compute contribution amounts** — Calculate each member's contribution due for the cycle. Lock amounts into a pending contribution ledger (immutable snapshot — this is the basis for all late fee calculations).
2. **Generate collection notices** — Notify all members of: amount due, due date, grace period end date, and each late fee tier threshold. Set status = `PENDING`.
3. **Snapshot available balances** — Record each member's cash account balance at cycle open. Used to determine whether auto-debit can partially or fully satisfy the contribution.

Phase 2 — Collection Window (Days 1 – N)

Auto-debit — full balance available

If Member Cash Account balance contribution due, post immediately on Day 1 (or the configured debit date).

Journal entry:

	Account	Amount
DR	Member Cash Account	Contribution amount
CR	Master Fund Account	Contribution amount
CR	Member Fund Account	Contribution amount (memo)

Status → `COLLECTED`

Partial debit — insufficient balance

If balance < contribution due: debit available balance, mark remaining shortfall as `PARTIALLY PENDING`. Continue monitoring for incoming deposits.

Journal entry:

	Account	Amount
DR	Member Cash Account	Available balance
CR	Master Fund Account	Available balance
CR	Member Fund Account	Partial amount (memo)

Status → `PARTIALLY PENDING`

Real-time deposit matching

Any deposit arriving during the window (bank import or direct cash deposit) triggers an immediate re-evaluation. If the shortfall is covered, post the remainder and mark `COLLECTED`.

> **Design note:** Every time a deposit posts to a member's cash account, the system should immediately check whether it closes an outstanding contribution shortfall — before the deposit becomes available for other uses. This prevents the gap where a member funds their account but the contribution remains flagged as overdue.

Phase 3 — Grace Period Close & Late Fee Engine (Day N+1 onward)

End-of-window sweep

At close of the collection window, flag all members with outstanding balances as `OVERDUE`. Record `overdue_since` timestamp — this is the immutable clock start for all late fee tier calculations.

> The `days_overdue` counter must start from the exact close-of-business timestamp of the collection window, not the cycle start date. Disputes almost always center on this boundary.

Tiered late fee schedule

Days overdue	Status	Action
1 – 3	OVERDUE	Reminders only — no fee applied
After day 3	LATE — Tier 1	Apply Fee X — post to member cash account
After day 10	LATE — Tier 2	Apply Fee Y (replaces Tier 1) — post to member cash account
After day 20	LATE — Tier 3	Apply Fee Z (replaces Tier 2) — escalate, flag account

Fee model options:

- **Replacement model (standard):** Each new tier replaces the prior fee. Tier 2 reverses Tier 1 and posts Tier 2.

- **Cumulative model:** Each new tier is added on top of prior fees. Each tier posts an additional DR to the member's cash account.

Journal entry — late fee application:

	Account	Amount
DR	Member Cash Account	Late fee amount
CR	Late Fee Income Account	Late fee amount

Nightly batch algorithm

```
FOR each member WHERE status IN ('OVERDUE', 'LATE_T1', 'LATE_T2', 'LATE_T3'):
  days = today - overdue_since_date
  new_tier = lookup_tier(days)          // e.g. {3: T1, 10: T2, 20: T3}
  IF new_tier != member.current_tier:
    reverse_prior_fee_entry(member)     // if replacement model
    post_late_fee(member, new_tier.fee)
    update_status(member, new_tier.label)
    notify_member(member)
```

Phase 4 — Settlement & Bank Reconciliation (Ongoing)

Bank import path

Bank statement lines are imported and matched against pending transactions. On match, the Master Cash Account entry is cleared.

Journal entry:

	Account	Amount
DR	Master Bank Account	Deposit amount
CR	Master Cash Account	Deposit amount

Direct cash deposit path

Cash deposited directly is posted immediately to the member's Cash Account, then reconciled later with a matching bank import entry.

Journal entry:

	Account	Amount
DR	Master Cash Account	Deposit amount
CR	Member Cash Account	Deposit amount

Cycle close & carry-forward

At month-end, reconcile Master Bank / Master Cash / Master Fund balances. Any unresolved shortfalls carry into the next cycle with accumulated late fees and escalated status.

Phase 5 — Reporting & Audit Trail

Collection summary report

Per-member output: amount due, amount collected, outstanding balance, fees applied, current status. Exported for administrator review.

Immutable audit log

Every journal entry, status change, fee application, and deposit match is written to a tamper-evident log with timestamp and operator ID. Required for dispute resolution and regulatory compliance.

Member Status State Machine

```
PENDING
    balance covers → COLLECTED
    partial balance → PARTIALLY PENDING
         deposit covers shortfall → COLLECTED
         window closes → OVERDUE
    window closes (zero balance) → OVERDUE
         day 3+ → LATE Tier 1
               day 10+ → LATE Tier 2
                     day 20+ → LATE Tier 3 (escalate)

Any OVERDUE / LATE state:
    payment received → SETTLING → COLLECTED
```

Configuration (per cycle)

Keep these in a configuration table — not hardcoded — to allow adjustments without deployment:

Parameter	Description
`collection_window_days`	Length of the collection window (N days)
`tier_1_day_threshold`	Days overdue before Tier 1 fee applies (e.g. 3)
`tier_2_day_threshold`	Days overdue before Tier 2 fee applies (e.g. 10)
`tier_3_day_threshold`	Days overdue before Tier 3 fee applies (e.g. 20)
`fee_tier_1`	Fee amount X
`fee_tier_2`	Fee amount Y
`fee_tier_3`	Fee amount Z
`fee_model`	`replacement` or `cumulative`